Chapter four

Oscillators and Waveform Generators

4.1 Introduction

An **oscillator** is a circuit which produces a continuous, repeated, alternating waveform without any input. Oscillators basically convert unidirectional current flow from a DC source into an alternating waveform which is of the desired frequency, as decided by its circuit components.

An oscillator is defined as a mechanical or electronic device that works on the principles of oscillation: a periodic fluctuation between two things based on changes in energy. Computers, clocks, watches, radios, and metal detectors are among the many devices that use oscillators.

A clock pendulum is a simple type of mechanical oscillator. The most accurate timepiece in the world, the atomic clock, keeps time according to the oscillation within atoms. Electronic oscillators are used to generate signals in computers, wireless receivers and transmitters, and audio-frequency equipment, particularly music synthesizers. There are many types of electronic oscillators, but they all operate according to the same basic principle: an oscillator always employs a sensitive amplifier whose output is fed back to the input in phase. Thus, the signal regenerates and sustains itself. This is known as positive feedback.

How oscillators work

The basic principle behind the working of oscillators can be understood by analyzing the behavior of an LC tank circuit shown in Figure 1 below, which employs an inductor L and a completely pre-charged capacitor C as its components. Here, at first, the capacitor starts to discharge via the inductor, which results in the conversion of its electrical energy into the electromagnetic field, which can be stored in the inductor. Once the capacitor discharges completely, there will be no current flow in the circuit.



LC tank circuit

Practically, the **oscillators** are nothing but the amplifier circuits which are provided with a positive or regenerative feedback wherein a part of the output signal is fed back to the input (Figure 3). Here the amplifier consists of an amplifying active element which can be a transistor or an Op-Amp and the back-fed in-phase signal is held responsible to keep-up (sustain) the oscillations by making-up for the losses in the circuit.

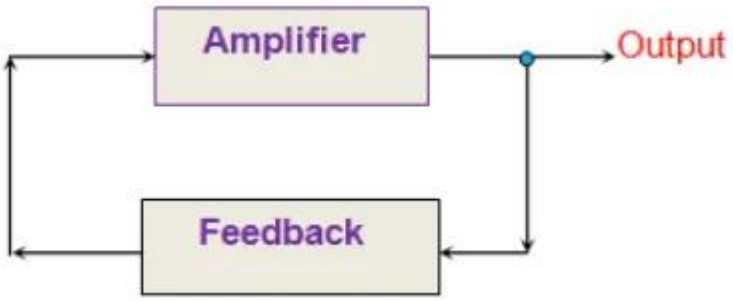


Figure 3 Typical oscillator

Once the power supply is switched ON, the oscillations will be initiated in the system due to the electronic noise present in it. This noise signal travels around the loop, gets amplified and converges to a single frequency sine wave very quickly. The expression for the closed-loop gain of the oscillator shown in Figure 3 is given as:

$$G = \frac{A}{1 + A\beta}$$

Where A is the voltage gain of the amplifier and β is the gain of the feedback network. Here, if $A\beta > 1$, then the oscillations will increase in amplitude (Figure 2a); while if $A\beta < 1$, then the oscillations will be damped (Figure 2b). On the other hand, $A\beta = 1$ leads to the oscillations which are of constant amplitude (Figure 2c). In other words, this indicates that if the feedback loop gain is small, then the oscillation dies-out, while if the gain of the feedback loop is large, then the output will be distorted; and only if the gain of feedback is unity, then the oscillations will be of constant amplitude leading to self-sustained oscillatory circuit.

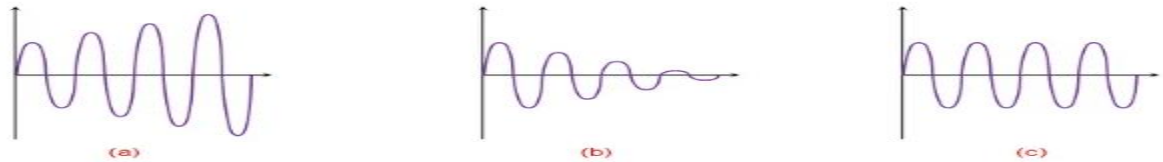


Figure 2 (a) Increasing Oscillations (b) Decaying Oscillations (c) Constant-Amplitude Oscillations

Type of Oscillator

Oscillators can be also be classified into various types depending on the parameter considered i.e. based on the feedback mechanism, the shape of the output waveform, etc.. These classifications types have been given below:

1. Classification Based on the Feedback Mechanism: Positive Feedback Oscillators and Negative Feedback Oscillators.
2. Classification Based on the Shape of the Output Waveform: Sine Wave Oscillators, Square or Rectangular Wave oscillators, Sweep Oscillators (which produce saw-tooth output waveform), etc.
3. Classification Based on the Frequency of the Output Signal: Low-Frequency Oscillators, Audio Oscillators (whose output frequency is of audio range), Radio Frequency Oscillators, High-Frequency Oscillators, Very High-Frequency Oscillators, Ultra High-Frequency Oscillators, etc.
4. Classification Based on the type of the Frequency Control Used: RC Oscillators, LC Oscillators, Crystal Oscillators (which use a quartz crystal to result in a frequency stabilized output waveform), etc.
5. Classification Based on the Nature of the Frequency of Output Waveform: Fixed Frequency Oscillators and Variable or Tunable Frequency Oscillators.

Oscillator Applications

Oscillators are a cheap and easy way to generate specific Frequency of a signal. For example, an RC oscillator is used to generate a Low Frequency signal, an LC oscillator is used to generate a High Frequency signal, and an Op-Amp based oscillator is used to generate a stable frequency.

The frequency of oscillation can be varied by varying the component value with potentiometer arrangements.

Some common applications of oscillators include:

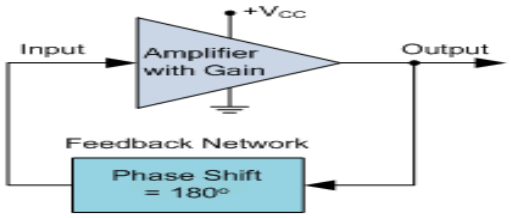
- Quartz watches (which uses a crystal oscillator)
- Used in various audio systems and video systems

- Used to generate clock pulses for microprocessors and micro-controllers
- Used in alarms and buzzes
- Used in metal detectors, stun guns, inverters, and ultrasonic
- Used to operate decorative lights (e.g. dancing lights)

4.2 Oscillators with RC feedback circuits

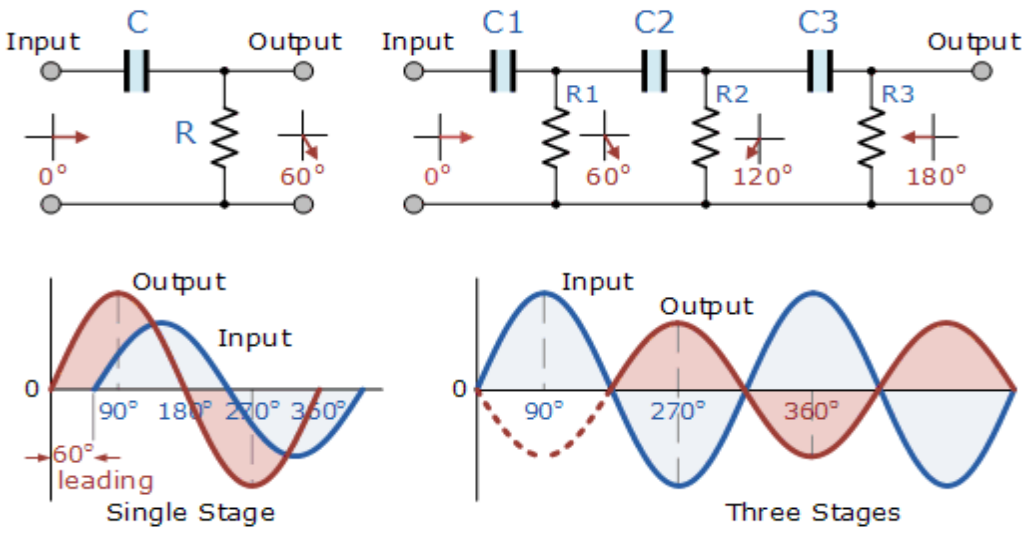
RC Oscillators use a combination of an amplifier and an RC feedback network to produce output oscillations due to the phase shift between the stages.

In an **RC Oscillator** circuit the input is shifted 180° through the feedback circuit returning the signal out-of-phase and 180° again through an inverting amplifier stage to produces the required positive feedback. This then gives us "180° + 180° = 360°" of phase shift which is effectively the same as 0°, thereby giving us the required positive feedback. In other words, the total phase shift of the feedback loop should be "0" or any multiple of 360° to obtain the same effect.



In a **Resistance-Capacitance Oscillator** or simply known as an **RC Oscillator**, we can make use of the fact that a phase shift occurs between the input to a RC network and the output from the same network by using interconnected RC elements in the feedback branch, for example.

RC Phase-Shift Network



The circuit on the left shows a single resistor-capacitor network whose output voltage "leads" the input voltage by some angle less than 90º. In a pure or ideal single-pole RC network, it would produce a maximum phase shift of exactly 90º, and because 180º of phase shift is required for oscillation, at least two single-poles networks must be used within an *RC oscillator* design.

However in reality it is difficult to obtain exactly 90º of phase shift for each RC stage so we must therefore use more RC stages cascaded together to obtain the required value at the oscillation frequency. The amount of actual phase shift in the circuit depends upon the values of the resistor (R) and the capacitor (C), at the chosen frequency of oscillations with the phase angle (φ) being given as:

RC Phase Angle

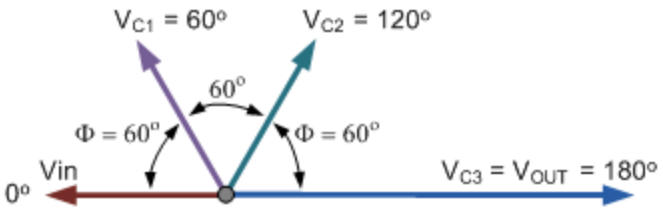
$$X_C = \frac{1}{2\pi f C} \qquad R = R,$$

$$Z = \sqrt{R^2 + (X_C)^2}$$

$$\therefore \quad \phi = \tan^{-1}\frac{X_C}{R}$$

Where: X_C is the Capacitive Reactance of the capacitor, R is the Resistance of the resistor, and *f* is the Frequency. In our simple example above, the values of R and C have been chosen so that at the required frequency the output voltage leads the input voltage by an angle of about 60º. Then the phase angle between each successive RC section increases by another 60º giving a phase difference between the input and output of 180º (3 x 60º) as shown by the following vector diagram.

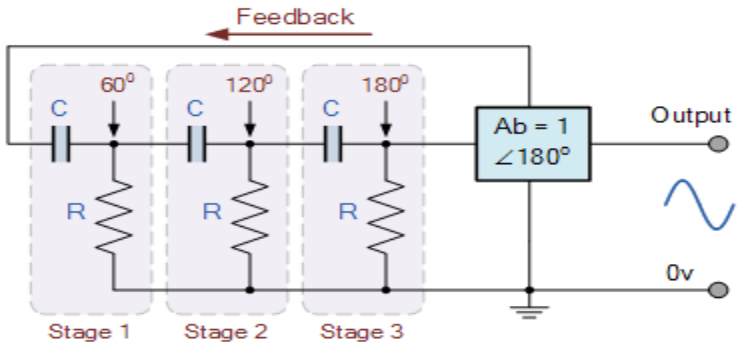
Vector Diagram



So by cascading together three such RC networks in series we can produce a total phase shift in the circuit of 180º at the chosen frequency and this form the bases of a "RC Oscillator" otherwise known as a **Phase Shift Oscillator** as the phase angle is shifted by an amount through each stage of the circuit.

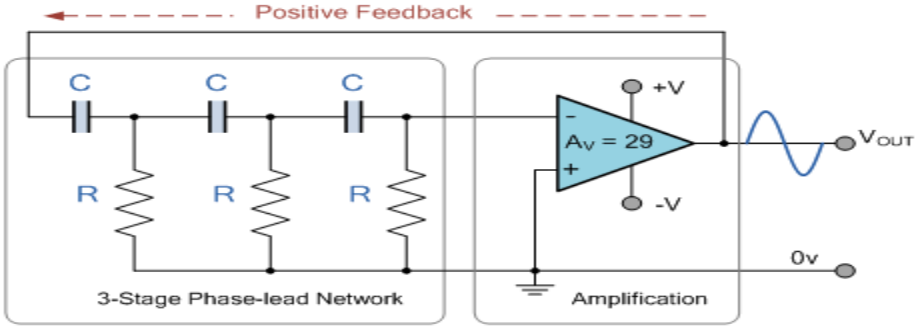
Then the phase shift occurs in the phase difference between the individual RC stages. Conveniently op-amp circuits are available in quad IC packages. For example, the LM124, or the LM324, etc. so four RC stages could also be used to produce the required 180° of phase shift at the required oscillation frequency.

We know that in an amplifier circuit either using a Bipolar Transistor or an Inverting Operational Amplifier configuration, it will produce a phase-shift of 180° between its input and output. If a three-stage RC phase-shift network is connected as a feedback network between the output and input of an amplifier circuit, then the total phase shift created to produce the required regenerative feedback is: 3 x 60° + 180° = 360° = 0° as shown.



The Op-amp RC Oscillator

When used as RC oscillators, **Operational Amplifier RC Oscillators** are more common than their bipolar transistors counterparts. The oscillator circuit consists of a negative-gain operational amplifier and a three section RC network that produces the 180° phase shift. The phase shift network is connected from the op-amps output back to its "inverting" input as shown below.

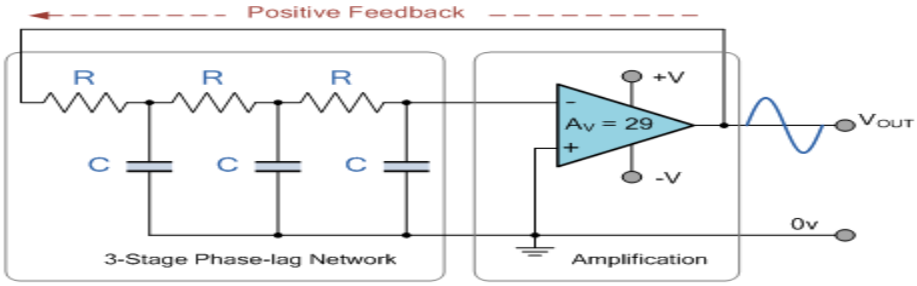


As the feedback is connected to the inverting input, the operational amplifier is therefore connected in its "inverting amplifier" configuration which produces the required 180° phase shift while the RC network produces the other 180° phase shift at the required frequency (180° + 180°).

This type of feedback connection with the capacitors in series and the resistors connected to ground (0V) potential is known as a *phase-lead* configuration. In other words, the output voltage leads the input voltage producing a positive phase angle.

But we can also create a *phase-lag* configuration by simply changing the positions of the RC components so that the resistors are connected in series and the capacitors are connected to ground (0V) potential as shown. This means that the output voltage lags the input voltage producing a negative phase angle.

Op-amp Phase-lag RC Oscillator Circuit



However, due to the reversal of the feedback components, the original equation for the frequency output of the phase-lead RC oscillator is modified to:

$$f_r = \frac{\sqrt{2N}}{2\pi RC}$$

Although it is possible to cascade together only two single-pole RC stages to provide the required 180º of phase shift (90º + 90º), the stability of the oscillator at low frequencies is generally poor. One of the most important features of an **RC Oscillator** is its frequency stability which is its ability to provide a constant frequency sine wave output under varying load conditions. By cascading three or even four RC stages together (4 x 45º), the stability of the oscillator can be greatly improved.

RC Oscillators with four stages are generally used because commonly available operational amplifiers come in quad IC packages so designing a 4-stage oscillator with 45º of phase shift relative to each other is relatively easy.

RC Oscillators are stable and provide a well-shaped sine wave output with the frequency being proportional to 1/RC and therefore, a wider frequency range is possible when using a variable capacitor. However, RC Oscillators are restricted to frequency applications because of their bandwidth limitations to produce the desired phase shift at high frequencies.

RC Oscillator Example No1

An operational amplifier based *3-stage RC Phase Shift Oscillator* is required to produce a sinusoidal output frequency of 4kHz. If 2.4nF capacitors are used in the feedback circuit, calculate the value of the frequency determining resistors and the value of the feedback resistor required to sustain oscillations. Also draw the circuit.

The standard equation given for the phase shift RC Oscillator is:

$$f_r = \frac{1}{2\pi RC\sqrt{2N}}$$

The circuit is to be a 3-stage RC oscillator which will therefore consist of equal resistors and three equal 2.4nF capacitors. As the frequency of oscillation is given as 4.0kHz, the value of the resistors are calculated as:

$$f = 4.0\text{kHz} = \frac{1}{2\pi\sqrt{(2\times 3)}\times R\times 2.4\text{nF}}$$

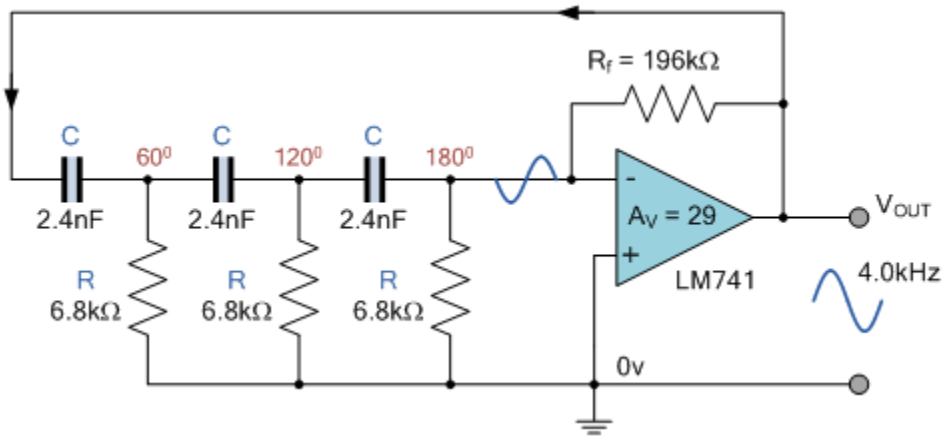
$$R = \frac{1}{2\pi\sqrt{(6)}\times 4000\times 2.4\times 10^{-9}}$$

The operational amplifiers gain must be equal to 29 in order to sustain oscillations. The resistive value of the oscillation resistors are 6.8kΩ, therefore the value of the op-amps feedback resistor R_f is calculated as:

$$A_V = \frac{R_F}{R} = 29$$

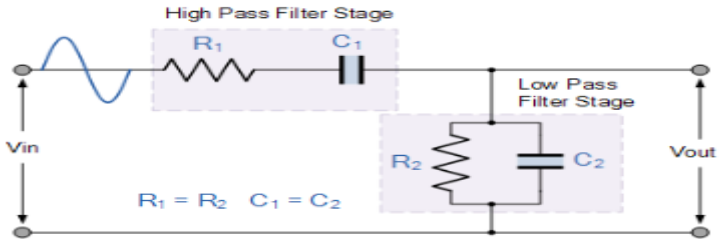
$$\therefore R_F = A_V \times R = 29\times 6.8\text{k}\Omega = 197.2\text{k}\Omega$$

RC Oscillator Op-amp Circuit



The Wien Bridge Oscillator

The Wien Bridge Oscillator uses two RC networks connected together to produce a sinusoidal oscillator.



In the *RC Oscillator* tutorial we saw that a number of resistors and capacitors can be connected together with an inverting amplifier to produce an oscillating circuit. One of the simplest sine wave oscillators which uses a RC network in place of the conventional LC tuned tank circuit to produce a sinusoidal output waveform, is called a **Wien Bridge Oscillator**.

The **Wien Bridge Oscillator** is so called because the circuit is based on a frequency-selective form of the Wheatstone bridge circuit. The Wien Bridge oscillator is a two-stage RC coupled amplifier circuit that has good stability at its resonant frequency, low distortion and is very easy to tune making it a popular circuit as an audio frequency oscillator but the phase shift of the output signal is considerably different from the previous phase shift **RC Oscillator**.

The **Wien Bridge Oscillator** uses a feedback circuit consisting of a series RC circuit connected with a parallel RC of the same component values producing a phase delay or phase advance circuit depending upon the frequency. At the resonant frequency $ƒr$ the phase shift is 0°. Consider the circuit below.

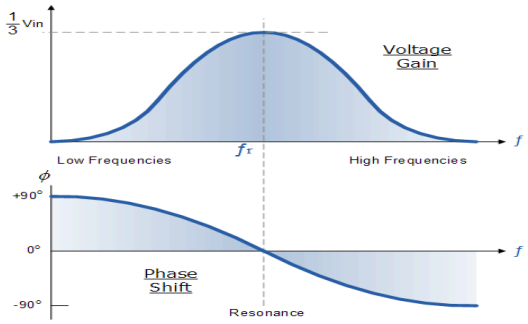
The above RC network consists of a series RC circuit connected to a parallel RC forming basically a High Pass Filter connected to a Low Pass Filter producing a very selective second-order frequency dependant Band Pass Filter with a high Q factor at the selected frequency, $ƒr$.

At low frequencies the reactance of the series capacitor (C1) is very high so acts a bit like an open circuit, blocking any input signal at Vin resulting in virtually no output signal, Vout. Likewise, at high frequencies, the reactance of the parallel capacitor, (C2) becomes very low, so this parallel connected capacitor acts a bit like a short circuit across the output, so again there is no output signal.

So there must be a frequency point between these two extremes of C1 being open-circuited and C2 being short-circuited where the output voltage, V_{OUT} reaches its maximum value. The frequency value of the input waveform at which this happens is called the oscillators *Resonant Frequency*, (fr).

At this resonant frequency, the circuit's reactance equals its resistance, that is: Xc = R, and the phase difference between the input and output equals zero degrees. The magnitude of the output voltage is therefore at its maximum and is equal to one third (1/3) of the input voltage as shown.

Oscillator Output Gain and Phase Shift



It can be seen that at very low frequencies the phase angle between the input and output signals is "Positive" (Phase Advanced), while at very high frequencies the phase angle becomes "Negative" (Phase Delay). In the middle of these two points the circuit is at its resonant frequency, (fr) with the two signals being "in-phase" or 0º. We can therefore define this resonant frequency point with the following expression.

Wien Bridge Oscillator Frequency

$$f_r = \frac{1}{2\pi RC}$$

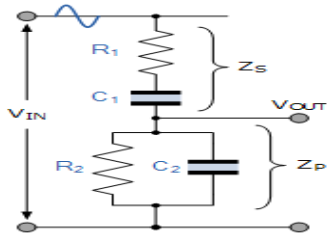
Where:

- fr is the Resonant Frequency in Hertz
- R is the Resistance in Ohms
- C is the Capacitance in Farads

We said previously that the magnitude of the output voltage, Vout from the RC network is at its maximum value and equal to one third (1/3) of the input voltage, Vin to allow for oscillations to occur. But why one third and not some other value.

In order to understand why the output from the RC circuit above needs to be one-third, that is 0.333xVin, we have to consider the complex impedance (Z = R ± jX) of the two connected RC circuits. We know from our AC Theory tutorials that the real part of the complex impedance is the resistance, R while the imaginary part is the reactance, X. As we are dealing with capacitors here, the reactance part will be capacitive reactance, Xc.

The RC Network



If we redraw the above RC network as shown, we can clearly see that it consists of two RC circuits connected together with the output taken from their junction. Resistor R_1 and capacitor C_1 form the top series network, while resistor R_2 and capacitor C_2 form the bottom parallel network.

Therefore the total DC impedance of the series combination (R_1C_1) we can call, Z_S and the total impedance of the parallel combination (R_2C_2) we can call, Z_P. As Z_S and Z_P are effectively connected together in series across the input, V_{IN}, they form a voltage divider network with the output taken from across Z_P as shown. Let's assume then that the component values of R_1 and R_2 are the same at: 12kΩ, capacitors C_1 and C_2 are the same at: 3.9nF and the supply frequency, f is 3.4kHz.

Series Circuit

The total impedance of the series combination with resistor, R_1 and capacitor, C_1 is simply:

$$R = 12k\Omega, \quad \text{but} \quad X_C = \frac{1}{2\pi fC}$$

$$\therefore X_C = \frac{1}{2\pi \times 3.4kHz \times 3.9nF} = 12k\Omega$$

$$Z_S = \sqrt{R^2 + X_C^2} = \sqrt{12000^2 + 12000^2}$$

$$\therefore Z_S = 16{,}970\Omega \text{ or } 17k\Omega$$

We now know that with a supply frequency of 3.4kHz, the reactance of the capacitor is the same as the resistance of the resistor at 12kΩ. This then gives us an upper series impedance Z_S of 17kΩ.

For the lower parallel impedance Z_P, as the two components are in parallel, we have to treat this differently because the impedance of the parallel circuit is influenced by this parallel combination.

Parallel Circuit

The total impedance of the lower parallel combination with resistor, R_2 and capacitor, C_2 is given as:

$$R = 12k\Omega, \quad \text{and} \quad X_C = 12k\Omega$$

$$\frac{1}{Z} = \frac{1}{R} + \frac{1}{XC} = \frac{1}{12000} + \frac{1}{12000}$$

$$\therefore Z = 6000\Omega \text{ or } 6k\Omega$$

At the supply frequency of 3400Hz, or 3.4kHz, the combined DC impedance of the RC parallel circuit becomes 6kΩ (R||Xc) with the vector sum of this parallel impedance being calculated as

$$R = 6k\Omega, \quad \text{and} \quad X_C = 6k\Omega \quad \text{(Parallel)}$$

$$Z_P = \sqrt{R^2 + X_C^2} = \sqrt{6000^2 + 6000^2}$$

$$\therefore Z_P = 8485\Omega \text{ or } 8.5k\Omega$$

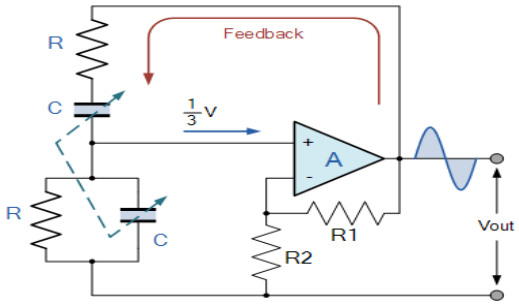
So we now have the value for the vector sum of the series impedance: 17kΩ, (Z_S = 17kΩ) and for the parallel impedance: 8.5kΩ, (Z_P = 8.5kΩ). Therefore the total output impedance, Zout of the voltage divider network at the given frequency is:

$$Z_{OUT} = \frac{Z_P}{Z_P + Z_S} = \frac{8.5k\Omega}{8.5k\Omega + 17k\Omega} = 0.333 \text{ or } \frac{1}{3}$$

Then at the oscillation frequency, the magnitude of the output voltage, Vout will be equal to Zout x Vin which as shown is equal to one third (1/3) of the input voltage, Vin and it is this frequency selective RC network which forms the basis of the **Wien Bridge Oscillator** circuit.

If we now place this RC network across a non-inverting amplifier which has a gain of 1+R1/R2 the following basic Wien bridge oscillator circuit is produced.

Wien Bridge Oscillator using op-amp



The output of the operational amplifier is fed back to both the inputs of the amplifier. One part of the feedback signal is connected to the inverting input terminal (negative or degenerative feedback) via the resistor divider network of R1 and R2 which allows the amplifiers voltage gain to be adjusted within narrow limits.

The other part, which forms the series and parallel combinations of R and C forms the feedback network and are fed back to the non-inverting input terminal (positive or regenerative feedback) via the RC Wien Bridge network and it is this positive feedback combination that gives rise to the oscillation.

The RC network is connected in the positive feedback path of the amplifier and has zero phase shift a just one frequency. Then at the selected resonant frequency, (fr) the voltages applied to the inverting and non-inverting inputs will be equal and "in-phase" so the positive feedback will cancel out the negative feedback signal causing the circuit to oscillate.

The voltage gain of the amplifier circuit MUST be equal too or greater than three "Gain = 3" for oscillations to start because as we have seen above, the input is 1/3 of the output. This value, ($Av \geq 3$) is set by the feedback resistor network, R1 and R2 and for a non-inverting amplifier this is given as the ratio 1+(R1/R2).

Also, due to the open-loop gain limitations of operational amplifiers, frequencies above 1MHz are unachievable without the use of special high frequency op-amps.

Wien Bridge Oscillator Example No1

Determine the maximum and minimum frequency of oscillations of a **Wien Bridge Oscillator** circuit having a resistor of 10kΩ and a variable capacitor of 1nF to 1000nF.

The frequency of oscillations for a Wien Bridge Oscillator is given as:

$$f_r = \frac{1}{2\pi RC}$$

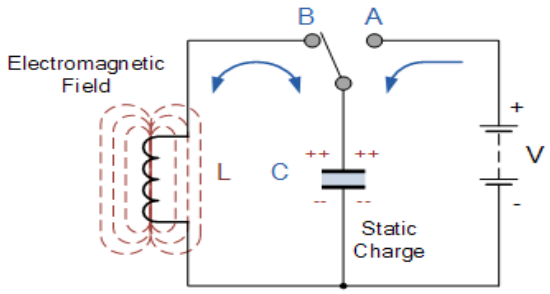
Wien Bridge Oscillator Lowest Frequency

$$f_{min} = \frac{1}{2\pi(10k\Omega)\times(1000\times10^{-9})} = 15.9\,Hz$$

LC Oscillator Basics

An *LC Oscillator* converts a DC input (the supply voltage) into an AC output (the waveform). This output waveform can have a wide range of different shapes and frequencies, and can be either complex in shape, or be a simple pure sine wave depending upon the application.

Oscillators are used in many pieces of test equipment producing either sinusoidal sine waves, square, sawtooth or triangular shaped waveforms or just a train of repetative pulses of a variable or constant width. LC Oscillators are commonly used in radio-frequency circuits because of their good phase noise characteristics and their ease of implementation.



The circuit consists of an inductive coil, L and a capacitor, C. The capacitor stores energy in the form of an electrostatic field and which produces a potential (*static voltage*) across its plates, while the inductive coil stores its energy in the form of an electromagnetic field. The capacitor is charged up to the DC supply voltage, V by putting the switch in position A. When the capacitor is fully charged the switch changes to position B.

The charged capacitor is now connected in parallel across the inductive coil so the capacitor begins to discharge itself through the coil. The voltage across C starts falling as the current through the coil begins to rise.

This rising current sets up an electromagnetic field around the coil which resists this flow of current. When the capacitor, C is completely discharged the energy that was originally stored in the capacitor, C as an electrostatic field is now stored in the inductive coil, L as an electromagnetic field around the coils windings.

As there is now no external voltage in the circuit to maintain the current within the coil, it starts to fall as the electromagnetic field begins to collapse. A back emf is induced in the coil ($e = -Ldi/dt$) keeping the current flowing in the original direction.

This current charges up capacitor, C with the opposite polarity to its original charge. C continues to charge up until the current reduces to zero and the electromagnetic field of the coil has collapsed completely.

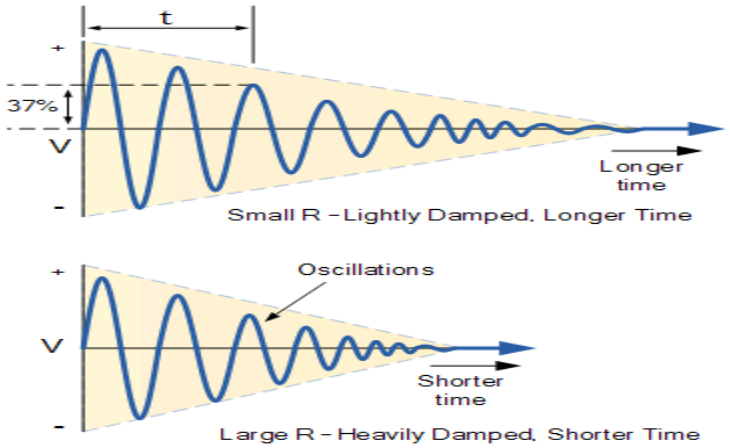
The energy originally introduced into the circuit through the switch, has been returned to the capacitor which again has an electrostatic voltage potential across it, although it is now of the opposite polarity. The capacitor now starts to discharge again back through the coil and the whole process is repeated. The polarity of the voltage changes as the energy is passed back and forth between the capacitor and inductor producing an AC type sinusoidal voltage and current waveform.

This process then forms the basis of an LC oscillators tank circuit and theoretically this cycling back and forth will continue indefinitely. However, things are not perfect and every time energy is transferred from the capacitor, C to inductor, L and back from L to C some energy losses occur which decay the oscillations to zero over time.

This oscillatory action of passing energy back and forth between the capacitor, C to the inductor, L would continue indefinitely if it was not for energy losses within the circuit. Electrical energy is lost in the DC or real resistance of the inductors coil, in the dielectric of the capacitor, and in radiation from the circuit so the oscillation steadily decreases until they die away completely and the process stops.

Then in a practical LC circuit the amplitude of the oscillatory voltage decreases at each half cycle of oscillation and will eventually die away to zero. The oscillations are then said to be “damped” with the amount of damping being determined by the quality or Q-factor of the circuit.

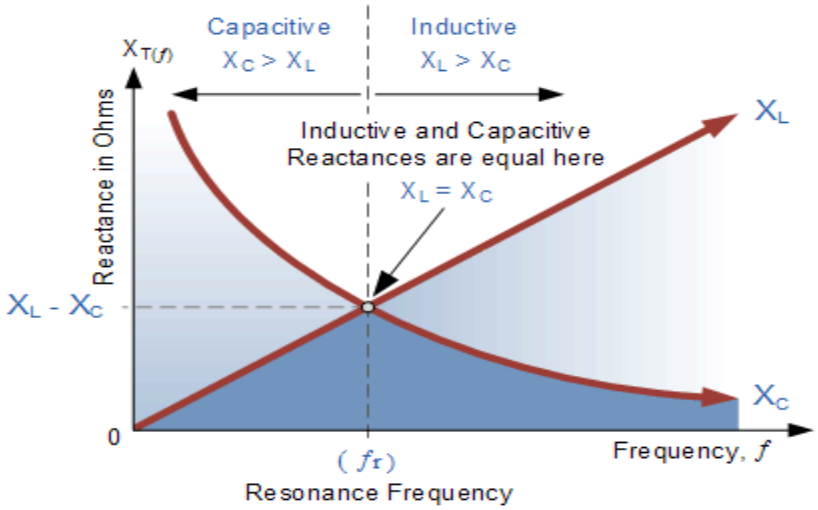
Damped Oscillations



The frequency of the oscillatory voltage depends upon the value of the inductance and capacitance in the LC tank circuit. We now know that for *resonance* to occur in the tank circuit, there must be a frequency point were the value of X_C, the capacitive reactance is the same as the value of X_L, the inductive reactance ($X_L = X_C$) and which will therefore cancel out each other out leaving only the DC resistance in the circuit to oppose the flow of current.

If we now place the curve for inductive reactance of the inductor on top of the curve for capacitive reactance of the capacitor so that both curves are on the same frequency axes, the point of intersection will give us the resonance frequency point, (f_r or ωr) as shown below.

Resonance Frequency



Where: f_r is in Hertz, L is in Henries and C is in Farads.

Then the frequency at which this will happen is given as:

$$X_L = 2\pi f L \quad \text{and} \quad X_C = \frac{1}{2\pi f C}$$

at resonance: $X_L = X_C$

$$\therefore \; 2\pi f L = \frac{1}{2\pi f C}$$

$$2\pi f^2 L = \frac{1}{2\pi C}$$

$$\therefore \; f^2 = \frac{1}{(2\pi)^2 LC}$$

$$f = \frac{\sqrt{1}}{\sqrt{(2\pi)^2 LC}}$$

Then by simplifying the above equation we get the final equation for **Resonant Frequency**, f_r in a tuned LC circuit as:

Resonant Frequency of a LC Oscillator

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

Where:

- L is the Inductance in Henries
- C is the Capacitance in Farads
- f_r is the Output Frequency in Hertz

This equation shows that if either L or C are decreased, the frequency increases. This output frequency is commonly given the abbreviation of (f_r) to identify it as the "resonant frequency".

To keep the oscillations going in an LC tank circuit, we have to replace all the energy lost in each oscillation and also maintain the amplitude of these oscillations at a constant level. The amount of energy replaced must therefore be equal to the energy lost during each cycle.

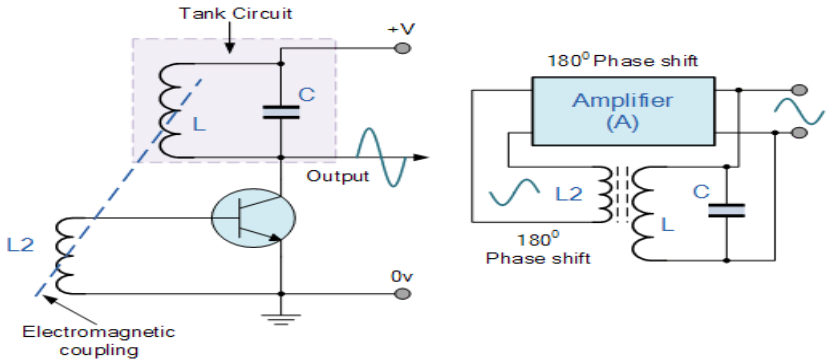
If the energy replaced is too large the amplitude would increase until clipping of the supply rails occurs. Alternatively, if the amount of energy replaced is too small the amplitude would eventually decrease to zero over time and the oscillations would stop.

The simplest way of replacing this lost energy is to take part of the output from the LC tank circuit, amplify it and then feed it back into the LC circuit again. This process can be achieved using a voltage amplifier using an op-amp, FET or bipolar transistor as its active device. However, if the loop gain of the feedback amplifier is too small, the desired oscillation decays to zero and if it is too large, the waveform becomes distorted.

To produce a constant oscillation, the level of the energy fed back to the LC network must be accurately controlled. Then there must be some form of automatic amplitude or gain control when the amplitude tries to vary from a reference voltage either up or down.

To maintain a stable oscillation the overall gain of the circuit must be equal to one or unity. Any less and the oscillations will not start or die away to zero, any more the oscillations will occur but the amplitude will become clipped by the supply rails causing distortion. Consider the circuit below.

Basic Transistor LC Oscillator Circuit



A Bipolar Transistor is used as the LC oscillators amplifier with the tuned LC tank circuit acts as the collector load. Another coil L2 is connected between the base and the emitter of the transistor whose electromagnetic field is "mutually" coupled with that of coil L.

"Mutual inductance" exists between the two circuits and the changing current flowing in one coil circuit induces, by electromagnetic induction, a potential voltage in the other (transformer effect) so as the oscillations occur in the tuned circuit, electromagnetic energy is transferred from coil L to coil L2 and a voltage of the same frequency as that in the tuned circuit is applied between the base and emitter of the transistor. In this way the necessary automatic feedback voltage is applied to the amplifying transistor.

The amount of feedback can be increased or decreased by altering the coupling between the two coils L and L2. When the circuit is oscillating its impedance is resistive and the collector and base voltages are 180° out of phase. In order to maintain oscillations (called frequency stability) the voltage applied to the tuned circuit must be "in-phase" with the oscillations occurring in the tuned circuit.

Therefore, we must introduce an additional 180° phase shift into the feedback path between the collector and the base. This is achieved by winding the coil of L2 in the correct direction relative to coil L giving us the correct amplitude and phase relationships for the **Oscillators** circuit or by connecting a phase shift network between the output and input of the amplifier.

The **LC Oscillator** is therefore a "Sinusoidal Oscillator" or a "Harmonic Oscillator" as it is more commonly called. LC oscillators can generate high frequency sine waves for use in radio frequency (RF) type applications with the transistor amplifier being of a Bipolar Transistor or FET.

Harmonic Oscillators come in many different forms because there are many different ways to construct an LC filter network and amplifier with the most common being the **Hartley LC Oscillator**, **Colpitts LC Oscillator**, **Armstrong Oscillator** and **Clapp Oscillator** to name a few.

LC Oscillator Example No1

An inductance of 200mH and a capacitor of 10pF are connected together in parallel to create an LC oscillator tank circuit. Calculate the frequency of oscillation.

$$f = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{200\text{mH} \times 10\text{pF}}} = 112.5\ \text{kHz}$$

Then we can see from the above example that by decreasing the value of either the capacitance, C or the inductance, L will have the effect of increasing the frequency of oscillation of the LC tank circuit.

LC Oscillators Summary

The basic conditions required for an **LC oscillator** resonant tank circuit are given as follows.

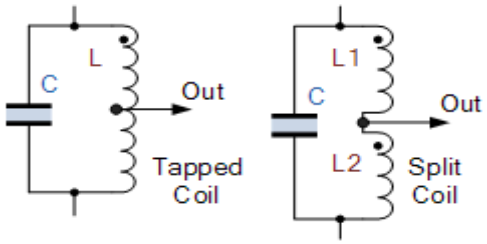
- For oscillations to exist an oscillator circuit MUST contain a reactive (frequency-dependant) component either an "Inductor", (L) or a "Capacitor", (C) as well as a DC power source.
- In a simple inductor-capacitor, LC circuit, oscillations become damped over time due to component and circuit losses.
- Voltage amplification is required to overcome these circuit losses and provide positive gain.
- The overall gain of the amplifier must be greater than one, unity.
- Oscillations can be maintained by feeding back some of the output voltage to the tuned circuit that is of the correct amplitude and in-phase, (0°).
- Oscillations can only occur when the feedback is "Positive" (self-regeneration).
- The overall phase shift of the circuit must be zero or 360° so that the output signal from the feedback network will be "in-phase" with the input signal.

The Hartley Oscillator

The Hartley Oscillator design uses two inductive coils in series with a parallel capacitor to form its resonance tank circuit producing sinusoidal oscillations.

In the **Hartley Oscillator** the tuned LC circuit is connected between the collector and the base of a transistor amplifier. As far as the oscillatory voltage is concerned, the emitter is connected to a tapping point on the tuned circuit coil.

The feedback part of the tuned LC tank circuit is taken from the centre tap of the inductor coil or even two separate coils in series which are in parallel with a variable capacitor, C as shown.



Hartley Oscillator Tank Circuit

Basic Hartley Oscillator Design

When the circuit is oscillating, the voltage at point X (collector), relative to point Y (emitter), is 180° out-of-phase with the voltage at point Z (base) relative to point Y. At the frequency of oscillation, the impedance of the Collector load is resistive and an increase in Base voltage causes a decrease in the Collector voltage.

Thus there is a 180° phase change in the voltage between the Base and Collector and this along with the original 180° phase shift in the feedback loop provides the correct phase relationship of positive feedback for oscillations to be maintained.

The amount of feedback depends upon the position of the "tapping point" of the inductor. If this is moved nearer to the collector the amount of feedback is increased, but the output taken between the Collector and earth is reduced and vice versa. Resistors, R1 and R2 provide the usual stabilizing DC bias for the transistor in the normal manner while the capacitors act as DC-blocking capacitors.

In this **Hartley Oscillator** circuit, the DC Collector current flows through part of the coil and for this reason the circuit is said to be "Series-fed" with the frequency of oscillation of the Hartley Oscillator being given as.

$$f = \frac{1}{2\pi\sqrt{L_T C}}$$

where: $L_T = L_1 + L_2 + 2M$

Note: L_T is the total cumulatively coupled inductance if two separate coils are used including their mutual inductance, M.

The frequency of oscillations can be adjusted by varying the "tuning" capacitor, C or by varying the position of the iron-dust core inside the coil (inductive tuning) giving an output over a wide range of frequencies making it very easy to tune. Also the **Hartley Oscillator** produces an output amplitude which is constant over the entire frequency range.

As well as the Series-fed Hartley Oscillator above, it is also possible to connect the tuned tank circuit across the amplifier as a shunt-fed oscillator as shown below.

Tutorial Example No1

A **Hartley Oscillator** circuit having two individual inductors of 0.5mH each, are designed to resonate in parallel with a variable capacitor that can be adjusted between 100pF and 500pF.

Determine the upper and lower frequencies of oscillation and also the Hartley oscillators bandwidth.

From above we can calculate the frequency of oscillations as being:

$$f_r = \frac{1}{2\pi\sqrt{L_T C}}$$

The circuit consists of two inductive coils in series, so the total inductance is given as:

$$L_T = L_1 + L_2 = 0.5\text{mH} + 0.5\text{mH} = 1.0\text{mH}$$

Upper Frequency Oscillation

$$f_H = \frac{1}{2\pi\sqrt{1\text{mH} \times 100\text{pF}}} = \frac{1}{6.283\sqrt{1\times10^{-13}}} = 503228\text{Hz}$$

$$\therefore f_H = 503\text{kHz}$$

Lower Frequency Oscillation

$$f_L = \frac{1}{2\pi\sqrt{1\text{mH} \times 500\text{pF}}} = \frac{1}{6.283\sqrt{5\times10^{-13}}} = 225050\text{Hz}$$

$$\therefore f_L = 225\text{kHz}$$

Hartley Oscillator Bandwidth

$$\text{Bandwidth} = f_H - f_L$$

$$= 503 - 225 = 278\text{kHz}$$

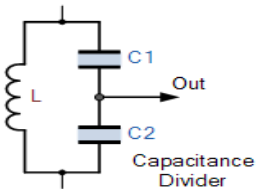
The Colpitts Oscillator

The Colpitts Oscillator design uses two centre-tapped capacitors in series with a parallel inductor to form its resonance tank circuit producing sinusoidal oscillations.

In many ways, the Colpitts oscillator is the exact opposite of the **Hartley Oscillator** we looked at in the previous tutorial. Just like the Hartley oscillator, the tuned tank circuit consists of an LC resonance sub-circuit connected between the collector and the base of a single stage transistor amplifier producing a sinusoidal output waveform.

The basic configuration of the **Colpitts Oscillator** resembles that of the *Hartley Oscillator* but the difference this time is that the centre tapping of the tank sub-circuit is now made at the junction of a "capacitive voltage divider" network instead of a tapped autotransformer type inductor as in the Hartley oscillator.

The Colpitts oscillator uses a capacitive voltage divider network as its feedback source. The two capacitors, C1 and C2 are placed across a single common inductor, L as shown. Then C1, C2 and L form the tuned tank circuit with the condition for oscillations being: $X_{C1} + X_{C2} = X_L$, the same as for the Hartley oscillator circuit.

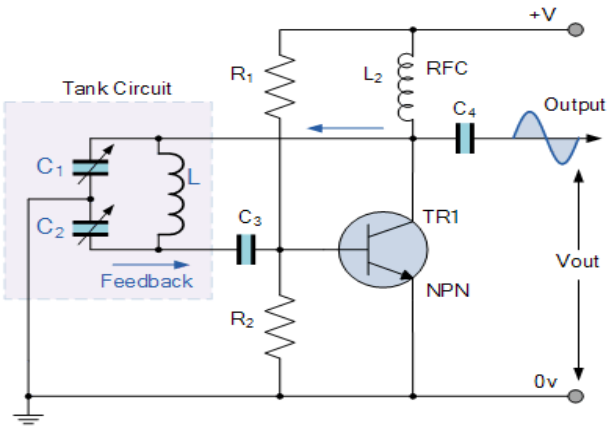


Colpitts Oscillator
Tank Circuit

The advantage of this type of capacitive circuit configuration is that with less self and mutual inductance within the tank circuit, frequency stability of the oscillator is improved along with a more simple design.

As with the Hartley oscillator, the Colpitts oscillator uses a single stage bipolar transistor amplifier as the gain element which produces a sinusoidal output. Consider the circuit below.

Basic Colpitts Oscillator Circuit



The emitter terminal of the transistor is effectively connected to the junction of the two capacitors, C1 and C2 which are connected in series and act as a simple voltage divider. When the power supply is firstly applied, capacitors C1 and C2 charge up and then discharge through the coil L. The oscillations across the capacitors are applied to the base-emitter junction and appear in the amplified at the collector output.

Resistors, R1 and R2 provide the usual stabilizing DC bias for the transistor in the normal manner while the additional capacitors act as a DC-blocking bypass capacitors. A radio-frequency choke (RFC) is used in the collector circuit to provide a high reactance (ideally open circuit) at the frequency of oscillation, (fr) and a low resistance at DC to help start the oscillations.

The required external phase shift is obtained in a similar manner to that in the Hartley oscillator circuit with the required positive feedback obtained for sustained undamped oscillations. The amount of feedback is determined by the ratio of C1 and C2. These two capacitances are generally "ganged" together to provide a constant amount of feedback so that as one is adjusted the other automatically follows.

The frequency of oscillations for a Colpitts oscillator is determined by the resonant frequency of the LC tank circuit and is given as:

$$f_r = \frac{1}{2\pi\sqrt{L\,C_T}}$$

where C_T is the capacitance of C1 and C2 connected in series and is given as:

$$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} \quad \text{or} \quad C_T = \frac{C_1 \times C_2}{C_1 + C_2}$$

The configuration of the transistor amplifier is of a *Common Emitter Amplifier* with the output signal 180° out of phase with regards to the input signal. The additional 180° phase shift require for oscillation is achieved by the fact that the two capacitors are connected together in series but in parallel with the inductive coil resulting in overall phase shift of the circuit being zero or 360°.

The amount of feedback depends on the values of C1 and C2. We can see that the voltage across C1 is the same as the oscillators output voltage, Vout and that the voltage across C2 is the oscillators feedback voltage. Then the voltage across C1 will be much greater than that across C2.

Therefore, by changing the values of capacitors, C1 and C2 we can adjust the amount of feedback voltage returned to the tank circuit. However, large amounts of feedback may cause the output sine wave to become distorted, while small amounts of feedback may not allow the circuit to oscillate.

Then the amount of feedback developed by the Colpitts oscillator is based on the capacitance ratio of C1 and C2 and is what governs the the excitation of the oscillator. This ratio is called the "feedback fraction" and is given simply as:

$$\text{Feedback Fraction} = \frac{C_1}{C_2}\,\%$$

Example No1

A **Colpitts Oscillator** circuit having two capacitors of 24nF and 240nF respectively are connected in parallel with an inductor of 10mH. Determine the frequency of oscillations of the circuit, the feedback fraction and draw the circuit.

The oscillation frequency for a Colpitts Oscillator is given as:

$$f_r = \frac{1}{2\pi\sqrt{L\,C_T}}$$

As the colpitts circuit consists of two capacitors in series, the total capacitance is therefore:

$$C_T = \frac{24\text{nF} \times 240\text{nF}}{24\text{nF} + 240\text{nF}} = 21.82\text{nF}$$

The inductance of the inductor is given as 10mH, then the frequency of oscillation is:

$$f_r = \frac{1}{2\pi\sqrt{LC_T}} = \frac{1}{6.283\sqrt{0.01 \times 21.82 \times 10^{-9}}} = 10.8\text{kHz}$$

The frequency of oscillations for the Colpitts Oscillator is therefore 10.8kHz with the feedback fraction given as:

$$F_F = \frac{C_1}{C_2} = \frac{24\text{nF}}{240\text{nF}} = 10\%$$

Colpitts Oscillator using an Op-amp (reading assignment +++)

Crystal oscillators

Crystal oscillators operate on the principle of inverse piezoelectric effect in which an alternating voltage applied across the crystal surfaces causes it to vibrate at its natural frequency. It is these vibrations which eventually get converted into oscillations.

These oscillators are usually made of Quartz crystal, even though other substances like Rochelle salt and Tourmaline exhibit the piezoelectric effect because, quartz is inexpensive, naturally-available and mechanically-strong when compared to others.

In crystal oscillators, the crystal is suitably cut and mounted between two metallic plates as shown by Figure 1a whose electrical equivalent is shown by Figure 1b. In reality, the crystal behaves like a series RLC circuit, formed by the components

1. A low-valued resistor R_S
2. A large-valued inductor L_S
3. A small-valued capacitor C_S

which will be in parallel with the capacitance of its electrodes C_p.

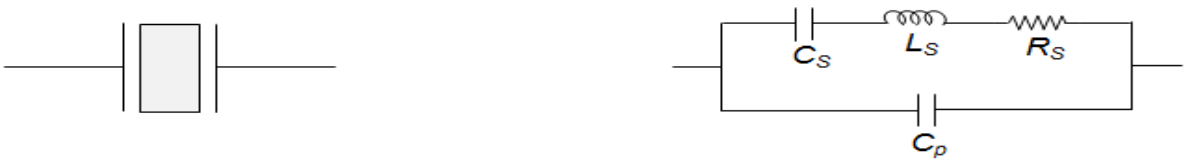


Figure 1 (a) Quartz Crystal (b) Equivalent Electric Circuit

Due to the presence of C_p, the crystal will resonate at two different frequencies viz.,

1. Series Resonant Frequency, f_s which occurs when the series capacitance C_S resonates with the series inductance L_S. At this stage, the crystal impedance will be the least and hence the amount of feedback will be the largest. Mathematical expression for the same is given as

$$f_s = \frac{1}{2\pi\sqrt{L_s C_s}}$$

2. Parallel Resonant frequency, f_p which is exhibited when the reactance of the L_SC_S leg equals the reactance of the parallel capacitor C_p i.e. L_S and C_S resonate with C_p. At this instant, the crystal impedance will be the highest and thus the feedback will be the least. Mathematically it can be given as

$$f_p = \frac{1}{2\pi\sqrt{L_S \frac{C_P C_S}{C_P + C_S}}}$$

The behavior of the capacitor will be capacitive both below f_S and above f_p. However for the frequencies which lie in-between f_S and above f_p, the crystal's behavior will be inductive. Further when the frequency becomes equal to parallel resonant frequency f_p, then the interaction between L_S and C_p would form a parallel tuned LC tank circuit. Hence, a crystal can be viewed as a combination of series and parallel tuned resonance circuits due to which one needs to tune the circuit for any one among these two. Moreover it is to be noted that f_p will be higher than f_s and the closeness between the two will be decided by the cut and the dimensions of the crystal in-use.

Crystal oscillators can be designed by connecting the crystal into the circuit such that it offers low impedance when operated in series-resonant mode (Figure 2a) and high impedance when operated in anti-resonant or parallel resonant mode (Figure 2b)

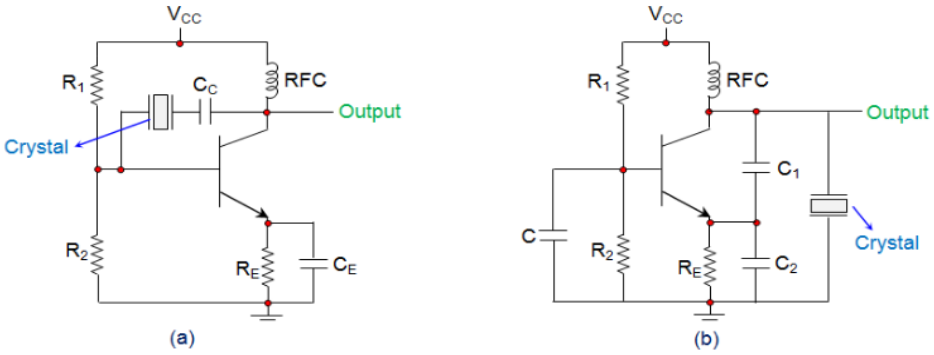


Figure 2 Crystal Oscillator Operating in (a) Series Resonance (b) Parallel Resonance

In the circuits shown, the resistors R_1 and R_2 form the voltage divider network while the emitter resistor R_E stabilizes the circuit. Further, C_E (Figure 2a) acts as an AC bypass capacitor while the coupling capacitor C_C (Figure 2a) is used to block DC signal propagation between the collector and the base terminals.

Next, the capacitors C_1 and C_2 form the capacitive voltage divider network in the case of Figure 2b. In addition, there is also a Radio Frequency Coil (RFC) in the circuits (both in Figure 2a and 2b) which offers dual advantage as it provides even the DC bias as well as frees the circuit-output from being affected by the AC signal on the power lines.

Crystal oscillators are compact in size and are of low cost due to which they are extensively used in electronic warfare systems, communication systems, guidance systems, microprocessors, microcontrollers, space tracking systems, measuring instruments, medical devices, computers, digital systems, instrumentation, phase-locked loop systems, modems, sensors, disk drives, marine systems, telecommunications, engine control systems, clocks, Global Positioning Systems (GPS), cable television systems, video cameras, toys, video games, radio systems, cellular phones, timers, etc.

Relaxation Oscillator

What is a Relaxation Oscillator?

A relaxation oscillator is defined as a non-linear electronic oscillator circuit that can generate a non-sinusoidal repetitive output signal. A relaxation oscillator was invented by Henri Abraham and Eugene Bloch using a vacuum tube during World War 1.

Oscillators are classified into two different categories; linear oscillators (for sinusoidal waveforms) and relaxation oscillators (for non-sinusoidal waveforms).

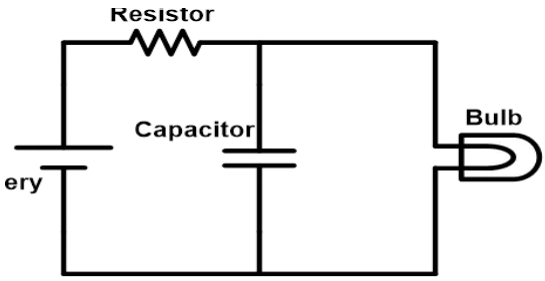
It must provide a repetitive and periodic signal for non-sinusoidal waveforms like triangular, square, and rectangular waves at its output.

To produce a cycle, the capacitor and inductor charge and discharge continuously. And the frequency of the cycle or period of oscillation depends on the time constant.

How Does a Relaxation Oscillator Work?

The relaxation oscillator contains energy storing devices like capacitor and inductor. These devices are charged by a source and discharge through a load.

The shape of the output waveform of the relaxation oscillator depends on the time constant of the circuit. Let’s understand the working of the relaxation oscillators with an example.



RC Relaxation Oscillator

Here, a capacitor is connected between a bulb and a battery. This circuit is also known as the flasher circuit or RC relaxation oscillator.

A battery charges the capacitor through the resistor. During charging of the capacitor, the bulb remains in OFF condition.

When the capacitor reaches its threshold value, it discharges through the bulb. Thus, during discharging of the capacitor, the bulb is glowing.

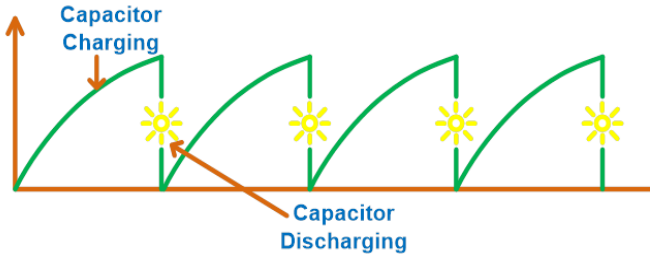
When the capacitor is discharged, it starts charging by the source again. And the bulb remains OFF.

So, the process of charging and discharging the capacitor is continuous and periodic.

The charging time of the capacitor determines by the time constant. And the time constant depends on the value of the resistor and capacitor for the RC circuit.

Therefore, the flashing rate of the bulb decides by the value of the resistor and capacitor.

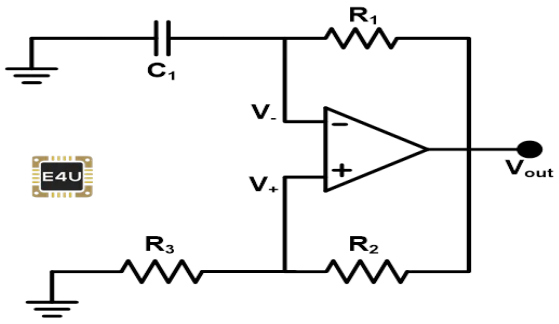
The waveforms across the bulb are as shown in the figure below.



RC Relaxation Oscillator Waveform

Op-Amp Relaxation Oscillator

An op-Amp relaxation oscillator is also known as an **astable multivibrator**. It is used to generate square waves. The circuit diagram of the Op-Amp relaxation oscillator is shown in the figure below.



Op-Amp Relaxation Oscillator

This circuit contains a capacitor, resistors, and an Op-Amp.

The non-inverting terminal of Op-Amp is connected with an RC circuit. So, the capacitor voltage V_C is the same as the voltage at non-inverting terminal V- of Op-Amp. And inverting terminal is connected with the resistors.

When the Op-Amp is used with positive feedback, as shown in the circuit diagram, the circuit is known as the Schmitt trigger. When V+ is greater than V-, the output voltage is +12V. And when V- is greater than V+, the output voltage is -12V.

For the initial condition, at time t=0, assume that the capacitor is fully discharged. Hence the voltage at the non-inverting terminal is V-=0. And the voltage at the inverting terminals V+ is equal to βV_{out}.

$$\beta = \frac{R_2}{R_2 + R_3}$$

To make calculation easy, we consider that R_2 and R_3 are the same. So, β=2 and βVout=6V. So, the capacitor will charge and discharge up to 6V.

$$t = 0; \quad V- = 0V; \quad V+ = +6V; \quad V_{OUT} = +12V$$

In this condition, V+ is greater than V-. So, output voltage V_{out}=+12V. And the capacitor starts charging.

When the capacitor voltage is greater than 6V, V- is greater than V+. Hence, the output voltage changes to -12V.

$$V- > 6V, \quad V+ = 6V, \quad V_{OUT} = -12V$$

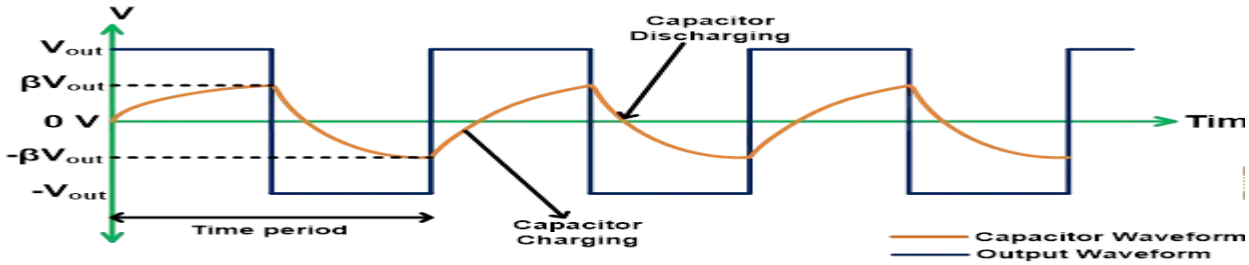
During this condition, the inverting terminal voltage changes its polarity. So, V+=-6V.

Now, the capacitor discharges up to -6V. When the capacitor voltage is less than -6V, again V+ is greater than V-.

$$V+ = -6V; \quad V- < -6V, \quad V+ > V-$$

Therefore, again the output voltage changes -12V to +12V. And again, the capacitor starts charging.

So, the charging and discharging cycle of the capacitor generates a periodic and repetitive square wave at the output terminal, as shown in the figure below.

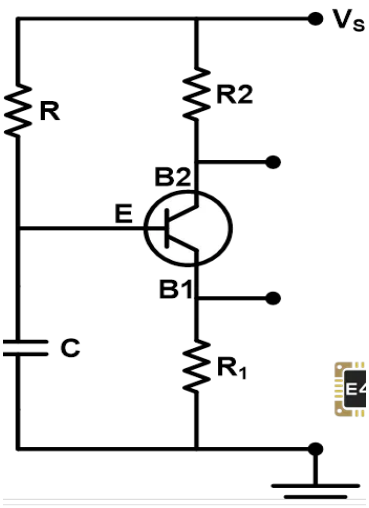


Op-Amp Relaxation Oscillator Waveform

The frequency of the output waveform depends on the charging and discharging time of the capacitor. And the charging-discharging time of the capacitor depends on the time constant of the RC circuit.

UJT Relaxation Oscillator

UJT (unijunction transistor) is used as a switching device in the relaxation oscillator. The circuit diagram of the UJT relaxation oscillator is shown in the figure below.



UJT Relaxation Oscillator

The emitter terminal of UJT is connected with a resistor and a capacitor.

We assume that initially the capacitor is discharged. So, the capacitor voltage is zero.

$$V_C = 0$$

In this condition, UJT remains OFF. And the capacitor starts charging through the resistor R by the equation below.

$$V = V_0(1 - e^{\frac{-t}{RC}})$$

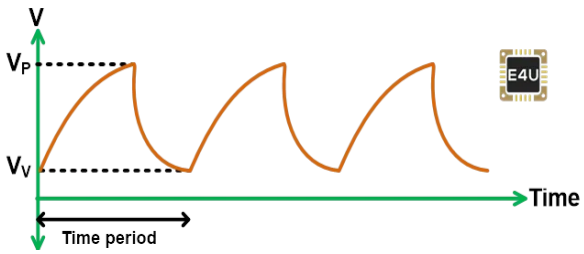
The capacitor continues to charge until it reaches the maximum supplied voltage V_{BB}.

When the voltage across the capacitor is greater than the supplied voltage, it enables the UJT turned ON. Then the capacitor stops charging and starts discharging through the resistor R_1.

The capacitor continues to discharge until the capacitor voltage reaches the valley voltage (V_V) of UJT. After that, the UJT turned OFF and started the capacitor charging.

Hence, the process of charging and discharging the capacitor generates a saw-tooth waveform across the capacitor. And the voltage appears across the resistor R_2 during discharging of the capacitor and remains zero during charging of the capacitor.

The voltage waveform across the capacitor and resistor R_2 is shown in the figure below



UJT Relaxation Oscillator Waveform

Relaxation Oscillator Frequency

The frequency of the Relaxation Oscillator depends on the charging and discharging time of the capacitor. In the RC circuit, the charging and discharging time decides by the time constant.

Frequency of Op-Amp Relaxation Oscillator

In the Op-Amp relaxation oscillator, the R_1 and C_1 contribute to the frequency of oscillation. Therefore, for lower frequency oscillation, we need a longer time for charging and discharging the capacitor. And for a long time of charging and discharging, we need to set a more significant R_1 and C_1.

Similarly, a smaller value of R_1 and C_1 causes higher frequency oscillation.

But, in the calculation of frequency, resistor R_2 and R_3 also play a vital role. Because these resistors will decide the threshold voltage of the capacitor, and the capacitor will charge up to this voltage level.

Suppose the threshold voltage is lower, the charging time is faster. Similarly, the threshold voltage is higher; the charging time is slower.

Therefore, the oscillation frequency depends on the value of R_1, R_2, R_3, and C_1. And Op-Amp relaxation oscillator frequency formula is;

$$f = \frac{1}{2 \times R_1 \times C_1 \times ln(\frac{1+k}{1-k})}$$

Where,

$$k = \frac{R_2}{R_2 + R_3}$$

In most of the conditions, R_2 and R_3 are the same to make easy design and calculation.

$$R_2 = R_3 = R$$

$$k = \frac{R}{2R} = \frac{1}{2}$$

$$f = \frac{1}{2 \times R_1 \times C_1 \times ln(\frac{1+\frac{1}{2}}{1-\frac{1}{2}})}$$

$$f = \frac{1}{2 \times R_1 \times C_1 \times ln(3)}$$

$$f = \frac{1}{2.2 \times R_1 \times C_1}$$

By putting the values of R_1 and C_1, we can find the oscillation frequency of the Op-Amp relaxation oscillator.

Frequency of UJT Relaxation Oscillator

In the UJT relaxation oscillator, also the frequency depends on the RC circuit. As shown in the circuit diagram of the UJT relaxation oscillator, resistors R_1 and R_2 are current limiting resistors. And the frequency of oscillation depends on the resistor R and capacitor C.

The formula of frequency for UJT relaxation oscillator is;

$$f = \frac{1}{RCln(\frac{1}{1-n})}$$

Where;

n = Intrinsic stand-off ratio. And the value of n lies between 0.51 to 0.82.

$$n = \frac{R_1}{R_1 + R_2}$$

To turn ON the UJT, minimum voltage required is;

$$V = nV_{BB} + V_D$$

Where,

V_{BB} = supply voltage

V_D = internal diode drop between emitter and base-2 terminal

The value of resistor R limits between following range.

$$max = \frac{V_{BB} - V_P}{I_P} \quad min = \frac{V_{BB} - V_V}{I_V}$$

Where,

V_P, I_P = peak voltage and current

V_V, I_V = valley voltage and current

555 Oscillator Tutorial

The 555 IC can be used to create a free running astable oscillator to continuously produce square wave pulses.

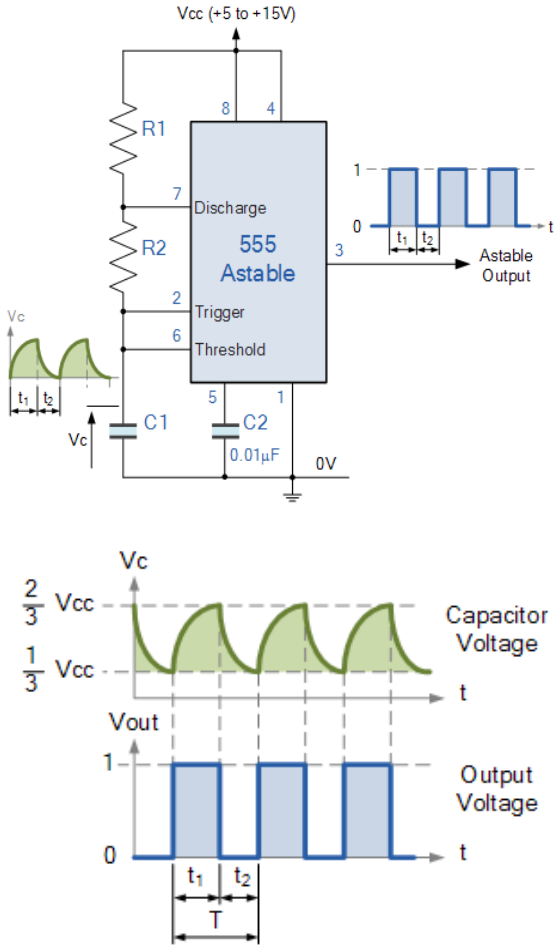
The **555 Oscillator** is another type of relaxation oscillator for generating stabilized square wave output waveforms of either a fixed frequency of up to 500kHz or of varying duty cycles from 50 to 100%. In the previous 555 Timer tutorial we saw that the Monostable circuit produces a single output one-shot pulse when triggered on its pin 2 trigger input.

Whereas the 555 monostable circuit stopped after a preset time waiting for the next trigger pulse to start over again, in order to get the 555 Oscillator to operate as an astable multivibrator it is necessary to continuously re-trigger the 555 IC after each and every timing cycle.

This re-triggering is basically achieved by connecting the *trigger* input (pin 2) and the *threshold* input (pin 6) together, thereby allowing the device to act as an astable oscillator. Then the 555 Oscillator has no stable states as it continuously switches from one state to the other. Also the single timing resistor of the previous monostable multivibrator circuit has been

split into two separate resistors, R1 and R2 with their junction connected to the *discharge* input (pin 7) as shown below.

Basic Astable 555 Oscillator Circuit



In the **555 Oscillator** circuit above, pin 2 and pin 6 are connected together allowing the circuit to re-trigger itself on each and every cycle allowing it to operate as a free running oscillator. During each cycle capacitor, C charges up through both timing resistors, R1 and R2 but discharges itself only through resistor, R2 as the other side of R2 is connected to the *discharge* terminal, pin 7.

Then the capacitor charges up to 2/3Vcc (the upper comparator limit) which is determined by the 0.693(R1+R2)C combination and discharges itself down to 1/3Vcc (the lower comparator limit) determined by the 0.693(R2*C) combination. This results in an output waveform whose voltage level is approximately equal to Vcc – 1.5V and whose output "ON" and "OFF" time

periods are determined by the capacitor and resistors combinations. The individual times required to complete one charge and discharge cycle of the output is therefore given as:

Astable 555 Oscillator Charge and Discharge Times

$$t_1 = 0.693(R_1 + R_2).C$$
and
$$t_2 = 0.693 \times R_2 \times C$$

Where, R is in Ω and C in Farads.

When connected as an astable multivibrator, the output from the 555 Oscillator will continue indefinitely charging and discharging between 2/3Vcc and 1/3Vcc until the power supply is removed. As with the monostable multivibrator these charge and discharge times and therefore the frequency are independent on the supply voltage.

The duration of one full timing cycle is therefore equal to the sum of the two individual times that the capacitor charges and discharges added together and is given as:

555 Oscillator Cycle Time

$$T = t_1 + t_2 = 0.693(R_1 + 2R_2).C$$

The output frequency of oscillations can be found by inverting the equation above for the total cycle time giving a final equation for the output frequency of an Astable 555 Oscillator as:

555 Oscillator Frequency Equation

$$f = \frac{1}{T} = \frac{1.44}{(R_1 + 2R_2).C}$$

By altering the time constant of just one of the RC combinations, the **Duty Cycle** better known as the "Mark-to-Space" ratio of the output waveform can be accurately set and is given as the ratio of resistor R2 to resistor R1. The Duty Cycle for the 555 Oscillator, which is the ratio of the "ON" time divided by the "OFF" time is given by:

555 Oscillator Duty Cycle

$$\text{Duty Cycle} = \frac{T_{ON}}{T_{OFF} + T_{ON}} = \frac{R_1 + R_2}{(R_1 + 2R_2)} \%$$

The duty cycle has no units as it is a ratio but can be expressed as a percentage (%). If both timing resistors, R1 and R2 are equal in value, then the output duty cycle will be 2:1 that is, 66% ON time and 33% OFF time with respect to the period.